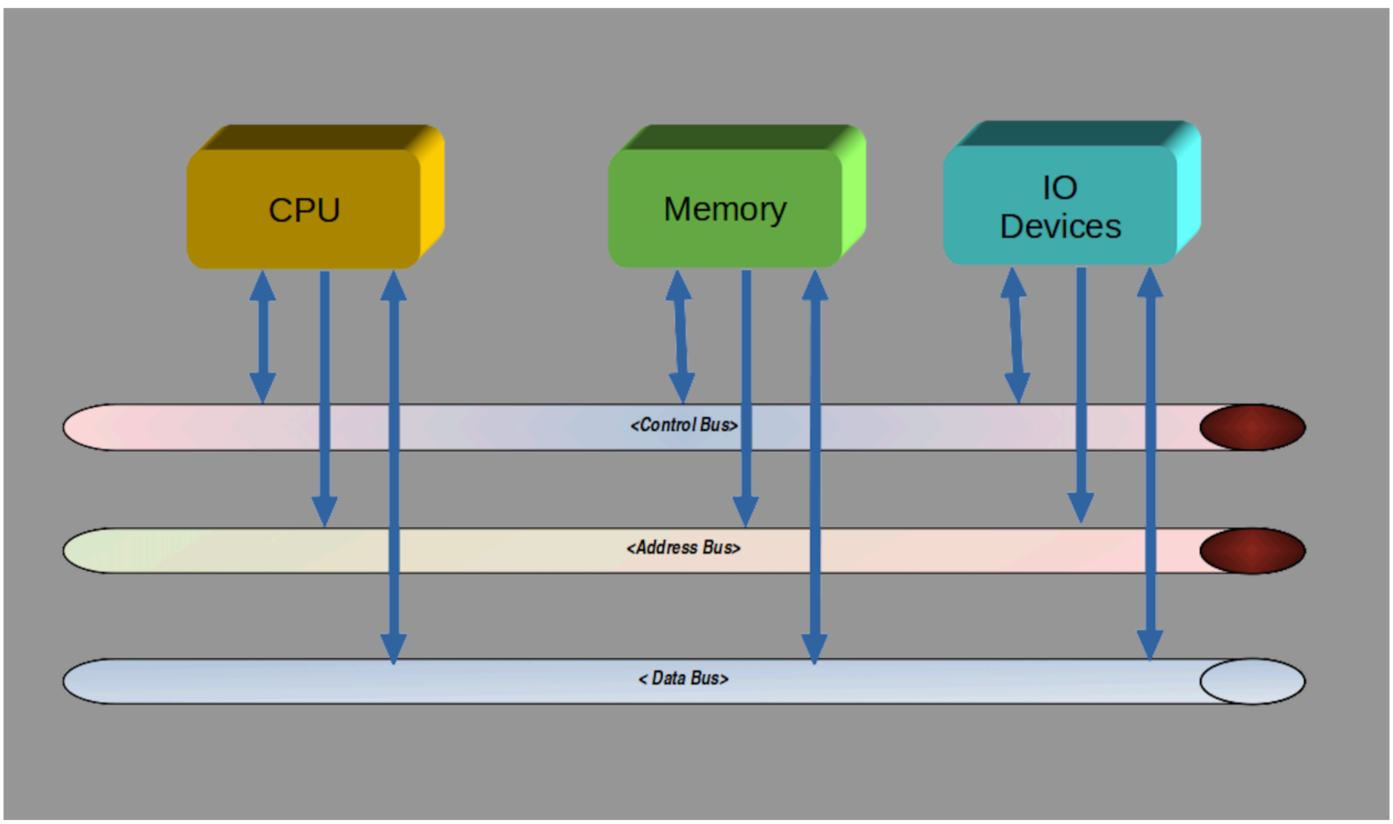
# Devices and Device Drivers

Device drivers control many different types of hardware. Standardization has helped define how hardware is assembled into “main boards” or “motherboards”, where additional components can be connected.

There are several “bus architectures” used to connect devices in a manner that the CPU can read, write and control the devices. Almost every bus has a selection method, an address mechanism and a data stream. Bus architectures have evolved over time, starting with simple designs with parallel data connections to complex multi-level designs to accommodate large complex implementations. The bus architecture is one of the elements in a computer where performance is observed. One of the benefits of standardization on interconnect methods (buses) is that there is an ever growing selection of devices to connect to a computer. USB is the most popular bus architecture.

USB connections are typically connected to intermediary hardware before being exposed to the Kernel drivers. These interface components are often arranged in a “bus” architecture. A selection signal may be sent to enable a bus member, allowing communication with the Kernel.




**Simple Bus Architecture**

A typical motherboard will have a collection of connections and buses depending on the motherboard form factor. There are decades of development and change in the bus architecture as improvements for speed and capacity continue to drive innovation. Additional reference information is included in the following short list of some of the common buses.

- **PCI**
  Peripheral Component Interconnect (PCI) is a local computer bus for attaching hardware devices to a computer and is part of the PCI Local Bus standard.
- **PCI Express**
  Peripheral Component Interconnect Express (PCI Express), abbreviated as PCIe or PCI-e, is a high-speed serial computer expansion bus standard, designed to replace the older PCI, PCI-X and AGP bus standards.